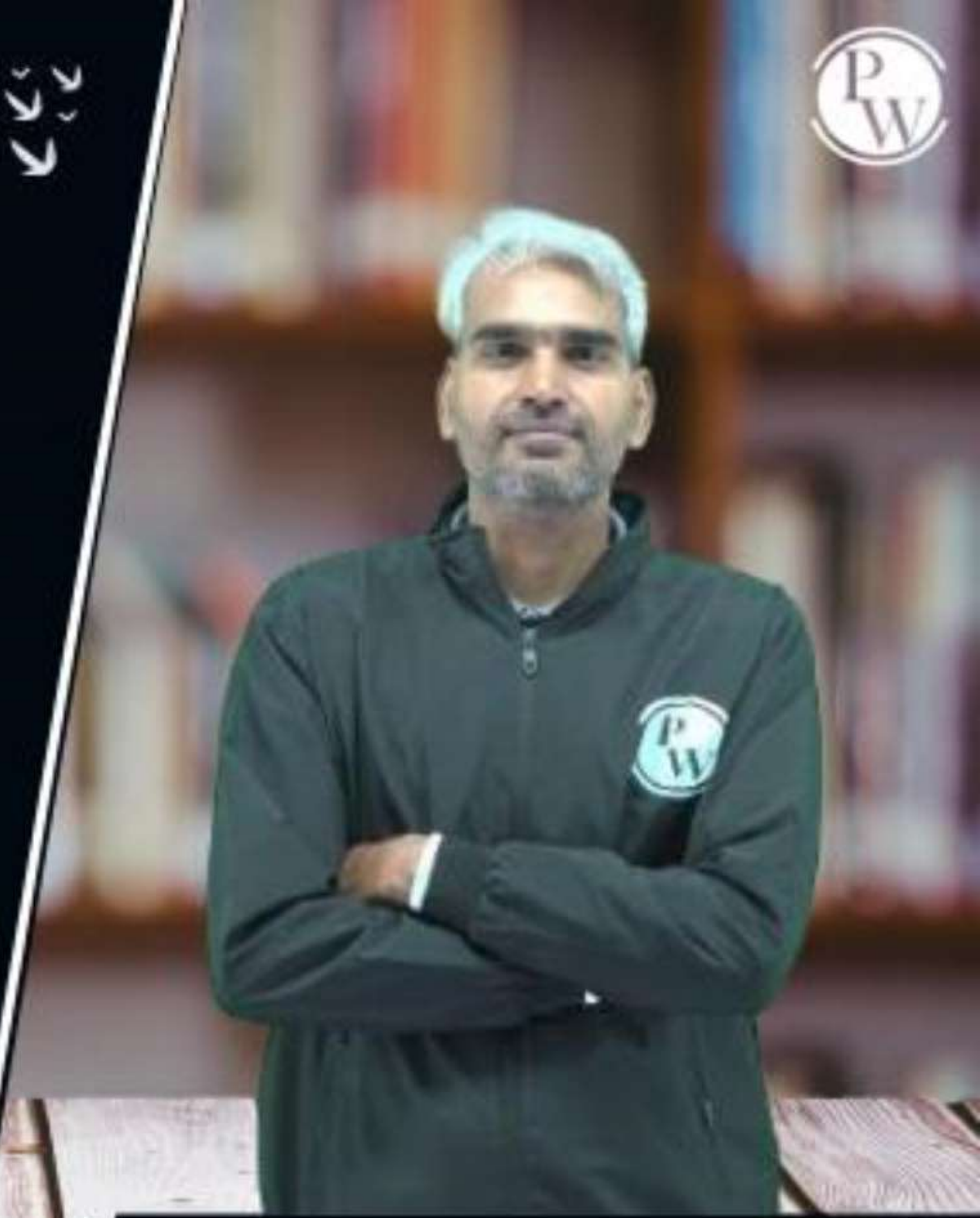
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 16



By- Vishal Sir

Recap of Previous Lecture

Topic Distributive lattice

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Topics to be Covered

Topic Boolean algebra / Boolean lattice

Topic Practice problems

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* Boolean Algebra / Boolean lattice

A lattice in which every element has Exactly one complement is called a Boolean lattice

A lattice which is* Complemented as well as distributive is called a Boolean lattice

Q:→ Which of the following statements is/ are not true

a) If A is any finite set then [P(A),⊆] is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice

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Q:- Which of the following statements is/ are not true

a) If A is any finite set then $[P(A), \subseteq]$ is distributive lattice (True)

b) Every sub lattice of a distributive lattice is also a distributive lattice

c) Every totally ordered set is a distributive lattice

d) Every totally ordered set is bounded

e) Every distributive lattice is bounded

f) Every distributive lattice is a complemented lattice

In P(A), w.r.t. relation $\subseteq$.
Every element of P(A) will have a unique complement; i.e. if $X \in P(A)$ then $\overline{X} = A - X$

$\therefore$ $(P(A), \subseteq)$ is complemented as well as distributive

let $A = \{1, 2, 3\}$

$P(A) = \{\underbrace{\phi}_{\text{Minimum}}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \underbrace{\{1,2,3\}}_{\text{Maximum}}\}$

$\overline{\phi} = \{1,2,3\}$, $\overline{\{3\}} = \{1,2\}$, $\overline{\{2,3\}} = \{1\}$

$\overline{\{1\}} = \{2,3\}$, $\overline{\{1,2\}} = \{3\}$, $\overline{\{1,2,3\}} = \phi$

$\overline{\{2\}} = \{1,3\}$, $\overline{\{1,3\}} = \{2\}$.

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Q:→ Which of the following statements is/ are not true

a) If A is any finite set then [P(A),⊆] is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice (True)
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice

Let $[M, \vee, \wedge]$ be a sub-lattice of lattice $[L, \vee, \wedge]$

if $a, b, c \in M$ then $\underbrace{a, b, c \in L \text{ as well}}$

We know 'L' is a distributive lattice

$\therefore a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

& $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

We know that lub & glb does not change for any pair of elements in sub-lattice

$\therefore$ same property will hold true in sub-lattice $[M, \vee, \wedge]$ as well

i.e.: If $[L, \vee, \wedge]$ is distributive then $[M, \vee, \wedge]$ is distributive

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Q:→ Which of the following statements is/ are not true

a) If A is any finite set then $[P(A), \subseteq]$ is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice { True } Already discussed
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice

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Q:→ Which of the following statements is/ are not true

a) If A is any finite set then [P(A),⊆] is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded {False} eg. $(N, \leq)$ it is a TOSET, but not bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice

other eg $(Z, \leq)$

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Q:- Which of the following statements is/ are not true

a) If A is any finite set then [P(A),⊆] is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded (False) : eg: $(N, \leq)$ is a TOSET it is not bounded but it is a distributive lattice
f) Every distributive lattice is a complemented lattice

Slide

Q:→ Which of the following statements is/ are not true

a) If A is any finite set then [P(A),⊆] is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice (False)

eg:- $(D_{12}, \div)$
it is distributive but not complemented
$\bar{2}$ = D.N.E.
$\bar{6}$ = D.N.E.

eg. $A = \{1,2,3\}$
$(A, \leq)$ is a TOSET
∴ distributive but
$\bar{2}$ = DNE

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Q:→ Which of the following statements is/ are not true

a) If A is any finite set then [P(A),⊆] is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
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d) Every totally ordered set is bounded
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* **Note:** → If 'n' is positive integer such that D_n has no element which is a perfect square except '1', then 'n' is called a square free integer

e.g. $D_{21} = \{\underbrace{1, 3, 7, 21}\}$ ↳ Except '1', none is a perfect square

Hence, 21 is a square free integer

Note: If n is a square free integer, then $(D_n, \div)$ is a Boolean lattice, and for an element $x \in D_n$, $\bar{x} = \frac{n}{x}$

Note: If n is a square free integer, then $(D_n, \div)$ is a Boolean lattice, and for an element $x \in D_n$, $\bar{x} = \frac{n}{x}$

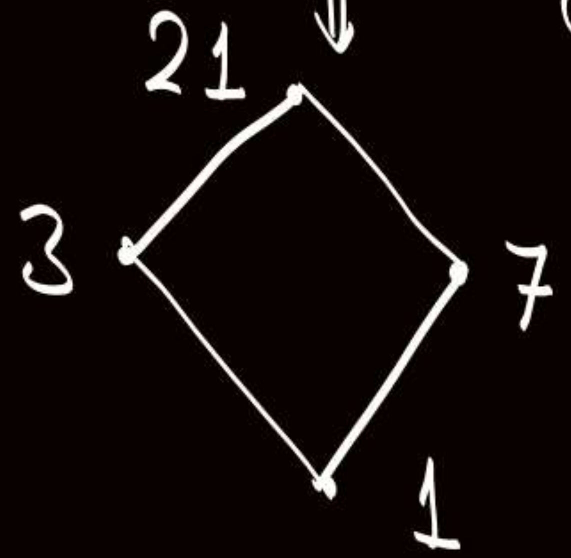
$\{1, 3, 7, 21\}$

e.g. $(D_{21}, \div)$

$\Downarrow$

Hasse diagram

$\Downarrow$



$\bar{1} = 21$

$\overline{21} = 1$

$\bar{3} = 7$

$\bar{7} = 3$

in lattice $(D_{21}, \div)$

"21" is a square free integer

$\therefore (D_{21}, \div)$ is a Boolean lattice.

and for $x \in D_{21}$, $\bar{x} = \frac{21}{x}$

$D_{21} = \{1, 3, 7, 21\}$

$\bar{1} = \frac{21}{1} = 21$, $\bar{3} = \frac{21}{3} = 7$, $\bar{7} = \frac{21}{7} = 3$, $\overline{21} = \frac{21}{21} = 1$

★ Which of the following lattice is /are not distributive?

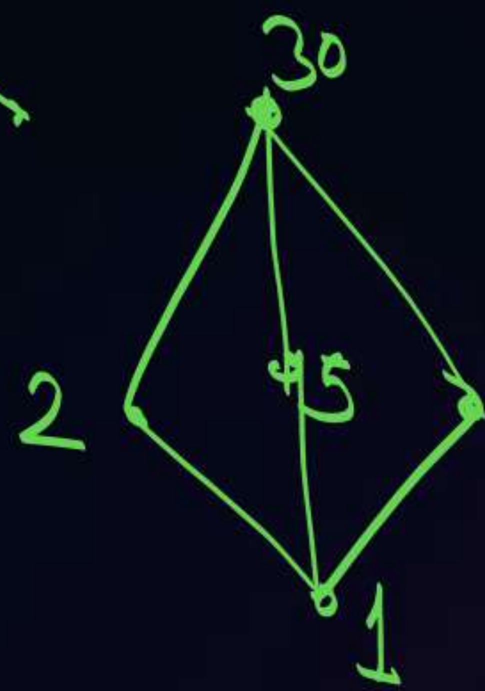
(A) $(D_{125}, \div)$ $\Rightarrow$ $D_{125} = \{1, 5, 25, 125\}$ $\Rightarrow$ A lattice of size ≤ 4 is always distributive

(B) $(P(A), \subseteq)$ $\Rightarrow$ distributive — it is a TOSET w.r.t. "÷" ∴ distributive

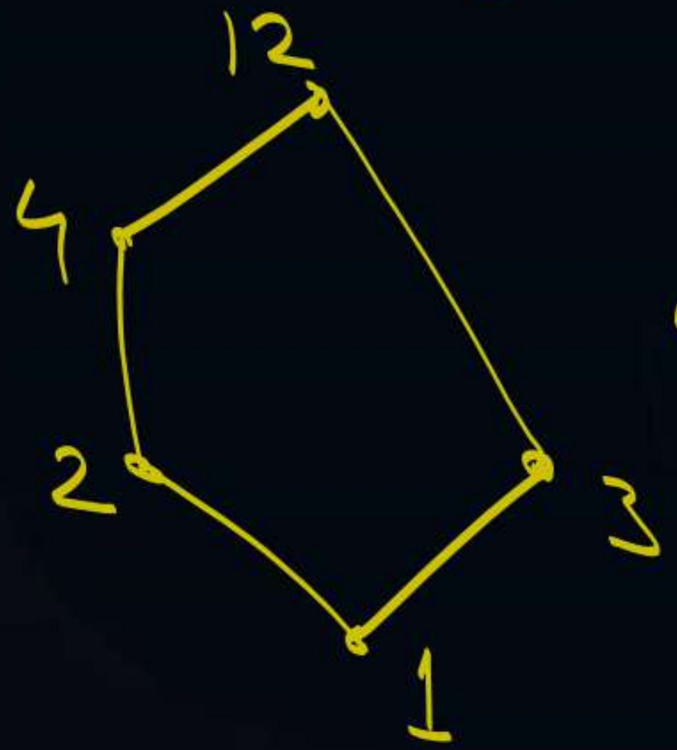
(C) $(D_{12}, \div)$ $\Longrightarrow$ No sublattice is isomorphic to L_1^* or L_2^* ∴ Distributive.

(D) $(\{1, 2, 3, 5, 30\}, \div)$

(E) $(\{1, 2, 3, 4, 12\}, \div)$ $\Rightarrow$



$\cong L_2^*$ ∴ Not distributive



$\cong L_1^*$ ∴ Not distributive.

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Which of the following lattice is /are Boolean Algebra?

(A) $(D_{125}, \div)$ ⇒ TOSET with '4' Elements ⇒ ∴ distributive, but not Complemented. } ∴ Not a Boolean Algebra

~~(B)~~ $(P(A), \subseteq)$ ⇒ for $X \in P(A)$ ⇒ $\overline{X} = A - X$

i.e. Every element has unique Complement ⇒ ∴ Distributive & Complemented

i.e. ∴ Boolean lattice

(C) $(D_{12}, \div)$ ⇒ $\overline{2}$ = D.N.E ⇒ ∴ Not Complemented

⟶ Hence, Not a Boolean lattice

(D) $(\{1, 2, 3, 5, 30\}, \div)$

(E) $(\{1, 2, 3, 4, 12\}, \div)$

$\cong L_1^*$ ∴ Not distributive

Hence, Not a Boolean lattice.

$\cong L_2^*$ ∴ Not distributive

Hence, Not a Boolean lattice

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Note :→ Boolean lattice

Lattice of size $= 2^0 = 1$

$\Downarrow$

$\Downarrow$

$\bar{a} = a$

$\Downarrow$

$\therefore$ It is a Boolean Algebra

Lattice of size $= 2^1 = 2$

$\bar{a} = b$

& $\bar{b} = a$

$\Downarrow$

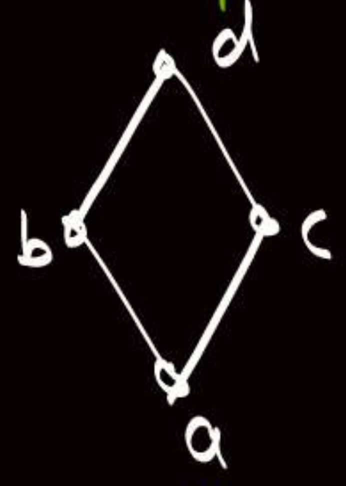
Every element has Exactly '1' Complement

$\therefore$ Boolean Algebra

Lattice of size $= 2^2 = 4$

Not complemented

$\therefore$ Not a Boolean Algebra



$\Downarrow$

$\bar{a} = d$

$\bar{d} = a$

$\bar{b} = c$

$\bar{c} = b$

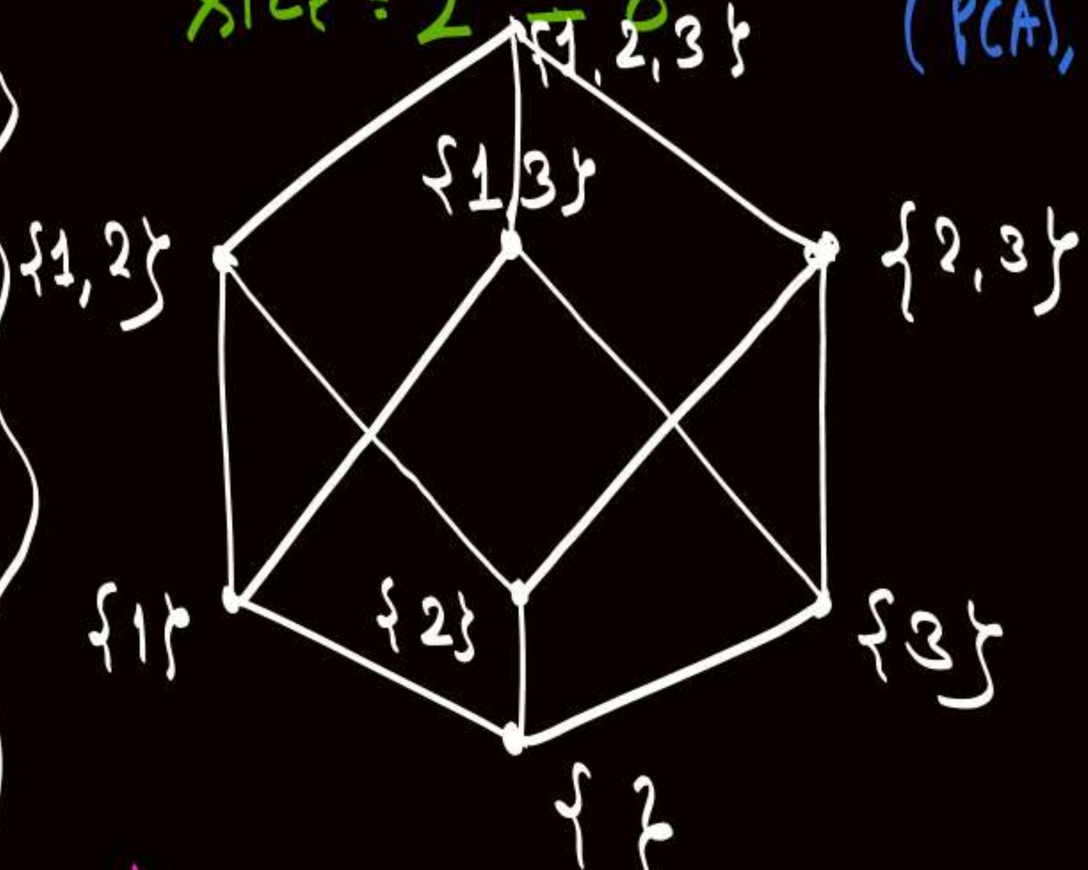
Every element has Exactly '1' Complement

$\therefore$ Boolean Algebra

Lattice of size $= 2^3 = 8$

let $A = \{1, 2, 3\}$

Lattice w.r.t. $(P(A), \subseteq)$



Unique Complement for Every element

$\therefore$ Boolean Algebra

Note:- ① If lattice is a Boolean Algebra then no. of Elements in lattice (Boolean Algebra) will always be in Power of '2'. {i.e; $2^0, 2^1, 2^2, \dots$} but Converse of the statement need not be true

"and"

If we have a Boolean algebra with 2^n elements, then number of Edges in Hasse diagram Corresponding to that Boolean Algebra will always be $n \cdot 2^{n-1}$

Q. let $U = \{1, 2, 3, 4, 5, 6, 7\}$

$A = P(U)$ $\quad |A| = 64$

Ex $B = \{\{1\}, \{2\}, \{2, 3\}\} \subseteq A$

let the no. of upper bounds that exist for the elements in set B is X and the no. of lower bount = y then "X+y" is.

$16 + 1 = 17$

Relation is "$\subseteq$"

$1 \times 1 \times 1 \times 2 \times 2 \times 2 \times 2 = 16$

$\{1, 2, 3, 4, 5, 6, 7\}$

the sets in A that are subset of all those element in set

"$\emptyset$" is the only set in A which is subset of all tree elements

$\therefore$ Only '1' lower bound.

2 mins Summary

Topic Boolean algebra / Boolean lattice

Topic Practice problems

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THANK - YOU

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